B. Divisor Subtraction

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1200 | Tags: implementation, math, number theory | https://codeforces.com/problemset/problem/1076/B

You are given an integer number n. The following algorithm is applied to it:

1. if $n = 0$, then end algorithm;
2. find the smallest **prime** divisor d of n;
3. subtract d from n and go to step 1.

Determine the number of subtrations the algorithm will make.

Input

The only line contains a single integer n ($2 \le n \le 10^{10}$).

Output

Print a single integer — the number of subtractions the algorithm will make.

Examples

Input
5

Output
1

Input
4

Output
2

Note

In the first example 5 is the smallest prime divisor, thus it gets subtracted right away to make a 0.

In the second example 2 is the smallest prime divisor at both steps.